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12 Summary: The Gate Taxonomy

#	Gate Name	Modality	Reversible?	Domain-Specific?	What It Enforces
1	Statistical	Linear algebra (eigenvalues)	Yes	Groups only	Inter-group correlation structure
2	Trigonometric	Angular embedding	Yes	Pairing choices	Closure / sum-to-one constraints
3	Geometric	Euclidean metrics	Partial	Vertex assignments	Balance / triangle inequality
4	Topological	Partitioning + revolution	No (bicameral)	Hierarchy rules	Democratic influence allocation
5	Electrochemical	Nonlinear scalar coupling	No	Coupling function	Physics-informed driving force
6	Recursive	Golden ratio recurrence	Yes	None	Uniform spacing / regularization
7	Neural Network	Commutative function spaces	Ill-conditioned	Training data	Distributional averaging
8	Equilibrium	Symmetric matrix mechanics	Partial	Normalization constants	Static force balance
9	Irreversibility	Computational thermodynamics	**No**	Reshape operation	Entropy generation / bounds
10	Integration	Elliptic special functions	Transcendental	Three input choices	Conservation by construction

Table 1: Gate Taxonomy: Each gate's modality, reversibility, domain-specific elements, and the constraint it enforces.

13 The Adaptability Matrix

For a new domain, the work required at each gate:

Gate	Effort	Adapt (domain-tuned)	Invariant (portable)
1. Statistical	**Med**	Groups; sampling scheme	CCA step
2. Trigonometric	**Low**	Pairings / angle map	$\sin^2 + \cos^2 = 1$ constraint
3. Geometric	**Low**	Vertex semantics	Metric/triangle invariants
4. Topological	**Med**	Partition/hierarchy rules	Spherization
5. Electrochemical	**High**	Nonlinear coupling (constitutive law)	Composite-variable scaffold
6. Recursive	**Zero**	General	Fibonacci / golden-ratio recurrence
7. Neural Network	**Low**	Training data	Commutative architecture
8. Equilibrium	**Low**	Scaling/normalization	Symmetry
9. Irreversibility	**Low**	Matrix size/reshape	SCT + Landauer bound logic
10. Integration	**Low**	Choose the 3 canonical inputs	AGM / elliptic-integral form

Table 2: Adaptability Matrix (lightweight): one-line adaptation vs. invariants for Gates 1–10.